Greedy algorithms

Problem 1-G-Coin Problem

Write a program to take value V and we want to make change for V Rs, and we have infinite supply of each of the denominations in Indian currency, i.e., we have infinite supply of { 1, 2, 5, 10, 20, 50, 100, 500, 1000} valued coins/notes, what is the minimum number of coins and/or notes needed to make the change.

Input Format:

Take an integer from stdin.

Output Format:

print the integer which is change of the number.

Example Input :

64

Output:

4

Explanaton:

We need a 50 Rs note and a 10 Rs note and two 2 rupee coins.

Source code:

```
#include <stdio.h>
int main() {
   int V;
   int denominations[] = {1, 2, 5, 10, 20, 50, 100, 500, 1000};
   int n = sizeof(denominations) / sizeof(denominations[0]);
   if (scanf("%d", &V) != 1) {
      return 0;
   }
   int total_count = 0;
   for (int i = n - 1; i >= 0; i--) {
      if (V >= denominations[i]) {
```

```
            total_count += V / denominations[i];
            V %= denominations[i];
        }
    }
    printf("%d\n", total_count);
    return 0;
}
```

Output:

Input	Expected	Got
49	5	5

Problem 2-G-Cookies Problem

Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.

Each child i has a greed factor g[i], which is the minimum size of a cookie that the child will be content with; and each cookie j has a size s[j]. If s[j] >= g[i], we can assign the cookie j to the child i, and the child i will be content. Your goal is to maximize the number of your content children and output the maximum number.

Example 1:

Input:

3

1 2 3

2

1 1

Output:

1

Explanation: You have 3 children and 2 cookies. The greed factors of 3 children are 1, 2, 3.

And even though you have 2 cookies, since their size is both 1, you could only make the child whose greed factor is 1 content.

You need to output 1.

Constraints:

1 <= g.length <= 3 * 10^4

0 <= s.length <= 3 * 10^4

1 <= g[i], s[j] <= 2^31 - 1

Source code:

```
#include <stdio.h>
#include <stdlib.h>
int compare(const void* a, const void* b) {
   return (*(int*)a - *(int*)b);
}
int findContentChildren(int* g, int gSize, int* s, int sSize) {
   qsort(g, gSize, sizeof(int), compare);
   qsort(s, sSize, sizeof(int), compare);
   int child = 0;
   int cookie = 0;
   while (child < gSize && cookie < sSize) {
      if (s[cookie] >= g[child]) {
         child++;
      }
      cookie++;
   }
   return child;
}
int main() {
   int gSize, sSize;
   if (scanf("%d", &gSize) != 1) return 0;
   int* g = (int*)malloc(gSize * sizeof(int));
```

```
    for (int i = 0; i < gSize; i++) {
        if (scanf("%d", &g[i]) != 1) return 0;
    }
    if (scanf("%d", &sSize) != 1) return 0;
    int* s = (int*)malloc(sSize * sizeof(int));
    for (int i = 0; i < sSize; i++) {
        if (scanf("%d", &s[i]) != 1) return 0;
    }
    int result = findContentChildren(g, gSize, s, sSize);
    printf("%d\n", result);
    free(g);
    free(s);
    return 0;
}
```

Output:

Input	Expected	Got
2 1 2 3 1 2 3	2	2

Problem 3-G-Burger Problem

A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs to run a distance to burn out his calories.

If he has eaten *i* burgers with c calories each, then he has to run at least $3^i * c$ kilometers to

burn out the calories. For example, if he ate 3
burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are $(3^0 * 1) + (3^1 * 3) + (3^2 * 2) = 1 + 9 + 18 = 28$.
But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine the minimum distance
he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy approach to solve the problem.

Input Format

First Line contains the number of burgers
Second line contains calories of each burger which is n space-separate integers

Output Format

Print: Minimum number of kilometers needed to run to burn out the calories

Sample Input

3
5 10 7

Sample Output

76

For example:

Test	Input	Result
Test Case 1	3 1 3 2	18

Source code:

```
#include <stdio.h>

#include <stdlib.h>


// Safe comparator to sort array in DESCENDING order

int compare(const void *a, const void *b) {
```

```
    long long arg1 = *(const long long *)a;
    long long arg2 = *(const long long *)b;

    if (arg1 > arg2) return -1;
    if (arg1 < arg2) return 1;
    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0;
    }

    long long *a = (long long *)malloc(n * sizeof(long long));
    for (int i = 0; i < n; i++) {
        scanf("%lld", &a[i]);
    }

    // Sort descending so largest calorie gets 3^0, second largest gets 3^1, etc.
    qsort(a, n, sizeof(long long), compare);

    unsigned long long total_distance = 0;
    unsigned long long power = 1;

    for (int i = 0; i < n; i++) {
        total_distance += a[i] * power;
        power *= 3;
```

```
    }

    printf("%llu\n", total_distance);

    free(a);
    return 0;
}
```

Output:

	Test	Input	Expected	Got	
✔	Test Case 1	3 1 3 2	18	18	✔
✔	Test Case 2	4 7 4 9 6	389	389	✔
✔	Test Case 3	3 5 10 7	76	76	✔

Problem 4-G-Array Sum max problem

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Source code:

```
#include <stdio.h>
#include <stdlib.h>
int compare(const void *a, const void *b) {
    return (*(int*)a - *(int*)b);
}
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int *arr = (int*)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    qsort(arr, n, sizeof(int), compare);
    long long max_sum = 0;
    for (int i = 0; i < n; i++) {
        max_sum += (long long)arr[i] * i;
    }
    printf("%lld\n", max_sum);
    free(arr);
    return 0;
}
```

Output:

Input	Expected	Got
5 2 5 3	40	40

Input	Expected	Got
4 0		
10 2 2 2 4 4 3 3 5 5 5	191	191
2 45 3	45	45

Problem 5-G-Product of Array elements-Minimum

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1	28

Input	Result
2 3 4 5 6	

Source code:

```
#include <stdio.h>
#include <stdlib.h>
int compareAsc(const void *a, const void *b) {
   return (*(int*)a - *(int*)b);
}
int compareDesc(const void *a, const void *b) {
   return (*(int*)b - *(int*)a);
}
int main() {
   int n;
   if (scanf("%d", &n) != 1) return 0;
   int *array_One = (int*)malloc(n * sizeof(int));
   int *array_Two = (int*)malloc(n * sizeof(int));
   for (int i = 0; i < n; i++) {
      scanf("%d", &array_One[i]);
   }
   for (int i = 0; i < n; i++) {
      scanf("%d", &array_Two[i]);
   }
```

```
    qsort(array_One, n, sizeof(int), compareAsc);
    qsort(array_Two, n, sizeof(int), compareDesc);
    long long min_sum = 0;
    for (int i = 0; i < n; i++) {
        min_sum += (long long)array_One[i] * array_Two[i];
    }
    printf("%lld\n", min_sum);
    free(array_One);
    free(array_Two);
    return 0;
}
```

Output:

Input	Expected	Got
3 1 2 3 4 5 6	28	28
4 7 5 1 2 1	22	22

Input	Expected	Got
3 4 1		
5 20 10 30 10 40 8 9 4 3 10	590	590